

NoisePrivacyMap<L2Distance<RBig>, MultiDP> for ZExpFamily<2>

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This proof resides in “contrib” because it has not completed the vetting process.

Proves soundness of the implementation of `NoisePrivacyMap` for `ZExpFamily<2>` with output measure `MultiDP`.

1 Hoare Triple

Precondition

Compiler-verified

`NoisePrivacyMap` is parameterized as follows:

- MI is `L2Distance<RBig>`.
- MO is `MultiDP`.

Caller-verified

None.

Pseudocode

```
1 class ZExpFamily2:
2     def noise_privacy_map(
3         self, _input_metric: L2Distance[RBig], _output_measure: MultiDP
4     ) -> PrivacyMap[L2Distance[RBig], MultiDP]:
5         zcdp_map = self.noise_privacy_map( #
6             L2Distance.default(), zCDP
7         )
8
9         def privacy_map(d_in: RBig):
10             rho = zcdp_map(d_in)
11             return PrivacyGuarantee().with_zCDP(rho, 0.0) #
12
13         return PrivacyMap.new_fallible(privacy_map)
```

Postcondition

Theorem 1.1. Given a distribution `self`, the implementation returns an error if `self` is invalid. Otherwise it returns a privacy map for the mechanism $\text{function}(\mathbf{x}) = \mathbf{x} + \mathbf{Z}$, where the coordinates of $\mathbf{Z}$ are independent discrete Gaussian samples represented by `self`, under `MultiDP`.

Proof. Line 5 delegates to the exact-zCDP privacy map proved in `NoisePrivacyMap_for_ZExpFamily2.tex`. That map validates the scale and sensitivity and returns the conservative value of ρ established by the discrete-Gaussian zCDP theorem.

Line 11 wraps this exact zCDP distance in a `PrivacyGuarantee` with source delta zero. Consequently the advertised `MultiDP` capability contains exactly the zCDP representation already proved for this mechanism. No Gaussian-DP or RDP representation is inferred from the distribution name; those representations, if useful, can be derived later by consumers using their respective theorems. The privacy guarantee is therefore valid under `MultiDP`, and increasing the distance to any `d_out` accepted by the measure ordering preserves the guarantee. $\square$

References